MEDIUM

ADVANCED MATH

Nonlinear functions

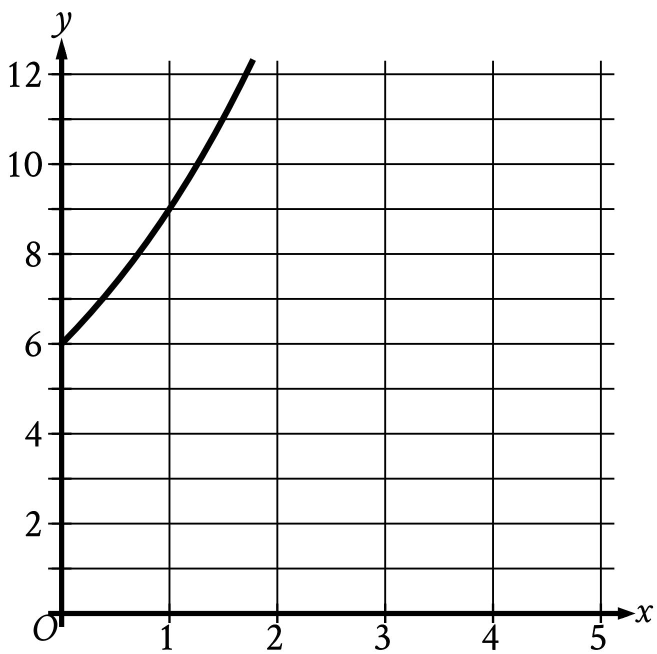
09

8 questions · ~12 min

SAT QUESTION BANK

2026-05-29

Q1 ADVANCED MATH



- Moving from left to right:
  - The curve passes through quadrant 1.
  - The curve trends up sharply.
- The curve passes through the following points:
  - (0 comma 6)
  - (1 comma 9.0)

The graph gives the estimated population y, in thousands, of a town x years since 2003, where $0 \leq x \leq 5$. Which of the following best describes the increase in the estimated population from $x = 0$ to $x = 1$?

A. The estimated population at $x = 1$ is 0.5 times the estimated population at $x = 0$.

B. The estimated population at $x = 1$ is 1.5 times the estimated population at $x = 0$.

C. The estimated population at $x = 1$ is 2.5 times the estimated population at $x = 0$.

D. The estimated population at $x = 1$ is 3.5 times the estimated population at $x = 0$.

Q2 ADVANCED MATH

$$q(x) = 32(2^x)$$

Which table gives three values of x and their corresponding values of $q(x)$ for function q?

A.

x	-1	0	1
$q(x)$	-64	0	64

B.

x	-1	0	1
$q(x)$	$\frac{1}{16}$	2	64

C.

x	-1	0	1
$q(x)$	$\frac{1}{16}$	32	64

D.

x	-1	0	1
$q(x)$	16	32	64

Q3 ADVANCED MATH

x	hx
0	1.23
2	1.54
4	1.94

The table shows the exponential relationship between the number of years, x, since Hana started training in pole vault, and the estimated height hx, in meters, of her best pole vault for that year. Which of the following functions best represents this relationship, where $x \leq 4$?

A. $hx = 1.120.23^x$

B. $hx = 1.121.23^x$

C. $hx = 1.230.12^x$

D. $hx = 1.231.12^x$

Q4

ADVANCED MATH

An object's kinetic energy, in joules, is equal to the product of one-half the object's mass, in kilograms, and the square of the object's speed, in meters per second. What is the speed, in meters per second, of an object with a mass of 4 kilograms and kinetic energy of 18 joules?

A. 3

B. 6

C. 9

D. 36

Q5

ADVANCED MATH

$$f(x) = (x+4)(x-1)(2x-3)$$

The function f is defined above. Which of the following is NOT an x-intercept of the graph of the function in the xy-plane?

A. $(-4, 0)$

B. $\left(-\frac{2}{3}, 0\right)$

C. $(1, 0)$

D. $\left(\frac{3}{2}, 0\right)$

Q6

ADVANCED MATH

The area of a triangle is 270 square centimeters. The length of the base of the triangle is 12 centimeters greater than the height of the triangle. What is the height, in centimeters, of the triangle?

A. 15

B. 18

C. 30

D. 36

Q7

ADVANCED MATH

A rectangular volleyball court has an area of 162 square meters. If the length of the court is twice the width, what is the width of the court, in meters?

A. 9

B. 18

C. 27

D. 54

Q8 ADVANCED MATH

x	gx
-1	25
0	1
1	$\frac{1}{25}$
2	$\frac{1}{625}$

For the exponential function g, the table shows four values of x and their corresponding values of gx. Which equation defines g?

A. $gx = -25^x$

B. $gx = -\frac{1}{25}^x$

C. $gx = 25^x$

D. $gx = \frac{1}{25}^x$